Practice Questions Date: *13th August, 2026*

1 What is the difference between STDIO and Streamable HTTP?

MCP (Model Context Protocol) allows an AI application or agent to communicate with external tools and services through an MCP server.

1. STDIO (Standard Input/Output)
With STDIO, the MCP client starts the MCP server as a local process. The client and server communicate through standard input and output streams.

Flow: AI Agent → MCP Client → STDIO → MCP Server → Tool/Data

Best for: local tools, learning, development, and desktop AI applications.

Example: The client starts a local server with a command such as “node server.js”.

2. Streamable HTTP
With Streamable HTTP, the MCP server runs as an HTTP service. The MCP client connects to it through an HTTP or HTTPS endpoint, so the server can be hosted separately from the AI application.

Flow: AI Agent → MCP Client → HTTP/HTTPS → MCP Server → Tool/Data

Best for: remote servers, cloud deployments, shared services, and multiple clients.

Example: An MCP server can be exposed through an endpoint such as “https://example.com/mcp”.

Key Differences

Feature	STDIO	Streamable HTTP
Connection	Local process streams	HTTP/HTTPS connection
Server location	Usually local	Local or remote
Setup	Simple	Needs HTTP/network setup
Multiple clients	Less convenient	Well suited
Remote access	No	Yes
Typical use	Local development	Production/shared services

Easy Way to Remember

STDIO = “Start the server on my machine and talk to it directly.”
Streamable HTTP = “Run the server as a web service and connect to it over HTTP.”